

Document Image Classification API Documentation

1. Introduction

This document describes the API developed for the Document Image Classification system. The API allows users to upload document images and obtain the predicted document category using a deep learning model.

2. API Overview

API Name: Document Image Classification API

Framework: FastAPI

Programming Language: Python

Model: Deep Learning Image Classification Model

3. API Endpoints

3.1 Health Check API

Method:

GET

Endpoint:

/

Purpose:

Checks whether the API is running.

Response:

```
{  
  "message": "Document Classification API is running"  
}
```

3.2 Document Prediction API

Method:

POST

Endpoint:

/predict

Input:

- Document image file (.jpg/.jpeg/.png)

Process:

1. Upload image
2. Resize image to 224×224
3. Normalize pixel values
4. Send image to trained model
5. Return predicted class and confidence score

Response Example:

```
{  
  "prediction": "scientific report",  
  "confidence": 19.92  
}
```

4. Supported Document Classes

- Handwritten
- Specification
- Email

- News Article
- Form
- Scientific Publication
- Questionnaire
- Presentation
- Memo
- Budget
- Advertisement
- Scientific Report
- Letter
- Invoice
- Resume
- File Folder

5. Technology Stack

- Python
- Fast API
- TensorFlow/Kera's
- NumPy
- HTML/CSS/JavaScript

6. Conclusion

The Document Image Classification API provides an interface between the trained deep learning model and the user interface. It enables automated classification of document images.